

A deterministic positive-weight counterexample to Conjecture 00000009876

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Claim and scope. The English statement asserts that passage-time variances of general finite-variance-weight first-passage percolation grow as the one-half power of distance, and that the exponent $1/2$ is universal under finite exponential moments. The Chinese statement makes the same universal assertions. Neither version requires a nonconstant edge-weight law. We disprove both assertions as written using strictly positive, independent, identically distributed weights on the standard square lattice. The example does not address a modified conjecture that explicitly excludes deterministic edge laws. A false universal clause suffices to disprove the conjunction; no claim concerning the heavy-tail clause is needed.

The probability model. Let $V = \mathbb{Z}^2$, and join two vertices when they differ by one in exactly one coordinate. Let E be the set of its undirected edges. On the one-point probability space $(\{*\}, \delta_*)$, set

$$W_e(*) = 1 \quad (e \in E).$$

Every W_e has law δ_1 , is strictly positive, has finite second moment, and satisfies $\mathbb{E}e^{tW_e} = e^t < \infty$ for every real t . These variables are independent, including in the full finite-family sense: for any finite $F \subset E$ and Borel sets B_e ,

$$\mathbb{P}\left(\bigcap_{e \in F} \{W_e \in B_e\}\right) = \prod_{e \in F} \mathbf{1}_{\{1 \in B_e\}} = \prod_{e \in F} \mathbb{P}(W_e \in B_e).$$

Thus this is an iid first-passage-percolation model, despite its degenerate distribution. In particular, neither lack of positivity nor lack of moments causes the counterexample.

Actual passage times. For a finite nearest-neighbor walk $p = (v_0, \dots, v_k)$, its cost is $C(p) = \sum_{j=0}^{k-1} W_{\{v_j, v_{j+1}\}} = k$. Define passage time by the usual infimum over all such walks:

$$T(x, y) = \inf\{C(p) : p \text{ is a finite nearest-neighbor walk from } x \text{ to } y\}.$$

Allowing walks with repeated vertices makes no difference here: all costs are positive. Put $T_n = T((n, 0), (0, 0))$, where $n \geq 0$ is an integer. Undirected symmetry also gives the same passage time in the reverse direction.

Lemma 1. *For every $n \geq 0$, $T_n = n$ at every outcome.*

Proof. Each edge can decrease the first coordinate by at most one. Thus a walk of length k from $(n, 0)$ to $(0, 0)$ must have $n \leq k$. More explicitly, the inequality $v_{j,1} \leq v_{j+1,1} + 1$ sums to $n \leq k$. Consequently every admissible cost is at least n . The horizontal walk $(n, 0), (n-1, 0), \dots, (0, 0)$ has exactly n edges and cost n . The set in the infimum is therefore nonempty and bounded below, its infimum equals n , and the infimum is attained. $\square$

Theorem 1. *The passage-time variances are identically zero. In particular there are no $c > 0$ and $N \in \mathbb{N}$ such that*

$$c\sqrt{n} \leq \text{Var}(T_n) \quad \text{for all } n \geq N.$$

Hence the universal one-half growth assertion is false.

Proof. The preceding lemma gives $\mathbb{E}T_n = n$ and $\text{Var}(T_n) = \mathbb{E}[(T_n - n)^2] = 0$. If the displayed inequality held, substituting $n = N + 1$ would give a strictly positive left side and zero right side, a contradiction. $\square$

Meaning of “grow as”. The contradiction applies to $\text{Var}(T_n) \sim c\sqrt{n}$ with $c > 0$, to two-sided comparability with $\sqrt{n}$, and to any assertion of a positive growth exponent $1/2$. Zero variance cannot have that exponent. An upper bound $O(\sqrt{n})$ alone would be consistent with this example, but would not assert growth as a one-half power or universality of that exponent, as the source does.

Formal verification. The accompanying Lean 4 project defines the square lattice as a `SimpleGraph` on $\mathbb{Z} \times \mathbb{Z}$, its actual undirected edge type, the Dirac probability measure, and the edge random variables. It proves their joint independence with Mathlib’s `iIndepFun`, their common law, positivity, second-moment integrability, and exponential-moment integrability for every real parameter. Costs are recursively summed along `SimpleGraph.Walk`; passage times use the real infimum of all walk costs. The coordinate lower bound and explicit horizontal walk establish `passage_eq_distance`. Mathlib’s measure-theoretic variance yields `passage_variance_zero`, followed by the explicit negation `no_positive_sqrt_lower_bound`. No asymptotic conclusion, shortest-path certificate, or variance identity is assumed as a hypothesis.